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compact1, compact2, compact3

java.util.concurrent

Interface ConcurrentNavigableMap<K,V>

Type Parameters:

`K - the type of keys maintained by this map`

`V - the type of mapped values`

All Superinterfaces:

`ConcurrentMap<K,V>, Map<K,V>, NavigableMap<K,V>, SortedMap<K,V>`

All Known Implementing Classes:

`ConcurrentSkipListMap`

```
public interface ConcurrentNavigableMap<K,V>
extends ConcurrentMap<K,V>, NavigableMap<K,V>
```

A `ConcurrentMap` supporting `NavigableMap` operations, and recursively so for its navigable sub-maps.

This interface is a member of the Java Collections Framework.

Since:

`1.6`

Nested Class Summary

Nested classes/interfaces inherited from interface java.util.Map

`Map.Entry<K,V>`

Method Summary

Abstract Methods

Modifier and Type	Method	Description
`NavigableSet<K>`	`descendingKeySet()`	Returns a reverse order `NavigableSet` view of the keys contained in this map.
`ConcurrentNavigableMap<K,V>`	`descendingMap()`	Returns a reverse order view of the mappings contained in this map.
`ConcurrentNavigableMap<K,V>`	`headMap(K toKey)`	Returns a view of the portion of this map whose keys are strictly less than `toKey`.
`ConcurrentNavigableMap<K,V>`	`headMap(K toKey, boolean inclusive)`	Returns a view of the portion of this map whose keys are less than (or equal to, if `inclusive` is true) `toKey`.
`NavigableSet<K>`	`keySet()`	Returns a `NavigableSet` view of the keys contained in this map.
`NavigableSet<K>`	`navigableKeySet()`	Returns a `NavigableSet` view of the keys contained in this map.

ConcurrentNavigableMap<K,V>	subMap(K fromKey, boolean fromInclusive, K toKey, boolean toInclusive)	Returns a view of the portion of this map whose keys range from fromKey to toKey.
ConcurrentNavigableMap<K,V>	subMap(K fromKey, K toKey)	Returns a view of the portion of this map whose keys range from fromKey, inclusive, to toKey, exclusive.
ConcurrentNavigableMap<K,V>	tailMap(K fromKey)	Returns a view of the portion of this map whose keys are greater than or equal to fromKey.
ConcurrentNavigableMap<K,V>	tailMap(K fromKey, boolean inclusive)	Returns a view of the portion of this map whose keys are greater than (or equal to, if inclusive is true) fromKey.

Methods inherited from interface java.util.concurrent.ConcurrentMap

compute, computeIfAbsent, computeIfPresent, forEach, getOrDefault, merge, putIfAbsent, remove, replace, replace, replaceAll

Methods inherited from interface java.util.NavigableMap

ceilingEntry, ceilingKey, firstEntry, floorEntry, floorKey, higherEntry, higherKey, lastEntry, lowerEntry, lowerKey, pollFirstEntry, pollLastEntry

Methods inherited from interface java.util.SortedMap

comparator, entrySet, firstKey, lastKey, values

Methods inherited from interface java.util.Map

clear, containsKey, containsValue, equals, get, hashCode, isEmpty, put, putAll, remove, size

Method Detail

subMap

```
ConcurrentNavigableMap<K,V> subMap(K fromKey,
                                   boolean fromInclusive,
                                   K toKey,
                                   boolean toInclusive)
```

Description copied from interface: NavigableMap

Returns a view of the portion of this map whose keys range from `fromKey` to `toKey`. If `fromKey` and `toKey` are equal, the returned map is empty unless `fromInclusive` and `toInclusive` are both true. The returned map is backed by this map, so changes in the returned map are reflected in this map, and vice-versa. The returned map supports all optional map operations that this map supports.

The returned map will throw an `IllegalArgumentException` on an attempt to insert a key outside of its range, or to construct a submap either of whose endpoints lie outside its range.

Specified by:

`subMap` in interface `NavigableMap<K,V>`

Parameters:

`fromKey` - low endpoint of the keys in the returned map

`fromInclusive` - true if the low endpoint is to be included in the returned view

`toKey` - high endpoint of the keys in the returned map

`toInclusive` - true if the high endpoint is to be included in the returned view

Returns:

a view of the portion of this map whose keys range from fromKey to toKey

Throws:

ClassCastException - if fromKey and toKey cannot be compared to one another using this map's comparator (or, if the map has no comparator, using natural ordering). Implementations may, but are not required to, throw this exception if fromKey or toKey cannot be compared to keys currently in the map.

NullPointerException - if fromKey or toKey is null and this map does not permit null keys

IllegalArgumentException - if fromKey is greater than toKey; or if this map itself has a restricted range, and fromKey or toKey lies outside the bounds of the range

headMap

```
ConcurrentNavigableMap<K,V> headMap(K toKey,
                                    boolean inclusive)
```

Description copied from interface: NavigableMap

Returns a view of the portion of this map whose keys are less than (or equal to, if `inclusive` is true) `toKey`. The returned map is backed by this map, so changes in the returned map are reflected in this map, and vice-versa. The returned map supports all optional map operations that this map supports.

The returned map will throw an `IllegalArgumentException` on an attempt to insert a key outside its range.

Specified by:

`headMap` in interface `NavigableMap<K,V>`

Parameters:

`toKey` - high endpoint of the keys in the returned map

`inclusive` - true if the high endpoint is to be included in the returned view

Returns:

a view of the portion of this map whose keys are less than (or equal to, if inclusive is true) toKey

Throws:

`ClassCastException` - if toKey is not compatible with this map's comparator (or, if the map has no comparator, if toKey does not implement `Comparable`). Implementations may, but are not required to, throw this exception if toKey cannot be compared to keys currently in the map.

`NullPointerException` - if toKey is null and this map does not permit null keys

`IllegalArgumentException` - if this map itself has a restricted range, and toKey lies outside the bounds of the range

tailMap

```
ConcurrentNavigableMap<K,V> tailMap(K fromKey,
                                    boolean inclusive)
```

Description copied from interface: NavigableMap

Returns a view of the portion of this map whose keys are greater than (or equal to, if `inclusive` is true) `fromKey`. The returned map is backed by this map, so changes in the returned map are reflected in this map, and vice-

versa. The returned map supports all optional map operations that this map supports.

The returned map will throw an `IllegalArgumentException` on an attempt to insert a key outside its range.

Specified by:

`tailMap` in interface `NavigableMap<K,V>`

Parameters:

`fromKey` - low endpoint of the keys in the returned map

`inclusive` - true if the low endpoint is to be included in the returned view

Returns:

a view of the portion of this map whose keys are greater than (or equal to, if inclusive is true) fromKey

Throws:

`ClassCastException` - if fromKey is not compatible with this map's comparator (or, if the map has no comparator, if fromKey does not implement `Comparable`). Implementations may, but are not required to, throw this exception if fromKey cannot be compared to keys currently in the map.

`NullPointerException` - if fromKey is null and this map does not permit null keys

`IllegalArgumentException` - if this map itself has a restricted range, and fromKey lies outside the bounds of the range

subMap

```
ConcurrentNavigableMap<K,V> subMap(K fromKey,
                                   K toKey)
```

Description copied from interface: `NavigableMap`

Returns a view of the portion of this map whose keys range from `fromKey`, inclusive, to `toKey`, exclusive. (If `fromKey` and `toKey` are equal, the returned map is empty.) The returned map is backed by this map, so changes in the returned map are reflected in this map, and vice-versa. The returned map supports all optional map operations that this map supports.

The returned map will throw an `IllegalArgumentException` on an attempt to insert a key outside its range.

Equivalent to `subMap(fromKey, true, toKey, false)`.

Specified by:

`subMap` in interface `NavigableMap<K,V>`

Specified by:

`subMap` in interface `SortedMap<K,V>`

Parameters:

`fromKey` - low endpoint (inclusive) of the keys in the returned map

`toKey` - high endpoint (exclusive) of the keys in the returned map

Returns:

a view of the portion of this map whose keys range from fromKey, inclusive, to toKey, exclusive

Throws:

`ClassCastException` - if fromKey and toKey cannot be compared to one another using this map's comparator (or, if the map has no comparator, using natural ordering). Implementations may, but are not required to, throw this exception if fromKey or toKey cannot be compared to keys currently in the map.

`NullPointerException` - if fromKey or toKey is null and this map does not permit null keys

`IllegalArgumentException` - if fromKey is greater than toKey; or if this map itself has a restricted range, and fromKey or toKey lies outside the bounds of the range

headMap

`ConcurrentNavigableMap<K,V> headMap(K toKey)`

Description copied from interface: `NavigableMap`

Returns a view of the portion of this map whose keys are strictly less than `toKey`. The returned map is backed by this map, so changes in the returned map are reflected in this map, and vice-versa. The returned map supports all optional map operations that this map supports.

The returned map will throw an `IllegalArgumentException` on an attempt to insert a key outside its range.

Equivalent to `headMap(toKey, false)`.

Specified by:

`headMap` in interface `NavigableMap<K,V>`

Specified by:

`headMap` in interface `SortedMap<K,V>`

Parameters:

`toKey` - high endpoint (exclusive) of the keys in the returned map

Returns:

a view of the portion of this map whose keys are strictly less than toKey

Throws:

ClassCastException - if toKey is not compatible with this map's comparator (or, if the map has no comparator, if toKey does not implement Comparable). Implementations may, but are not required to, throw this exception if toKey cannot be compared to keys currently in the map.

NullPointerException - if toKey is null and this map does not permit null keys

IllegalArgumentException - if this map itself has a restricted range, and toKey lies outside the bounds of the range

tailMap

ConcurrentNavigableMap<K,V> tailMap(K fromKey)

Description copied from interface: NavigableMap

Returns a view of the portion of this map whose keys are greater than or equal to fromKey. The returned map is backed by this map, so changes in the returned map are reflected in this map, and vice-versa. The returned map supports all optional map operations that this map supports.

The returned map will throw an IllegalArgumentException on an attempt to insert a key outside its range.

Equivalent to tailMap(fromKey, true).

Specified by:

tailMap in interface NavigableMap<K,V>

Specified by:

tailMap in interface SortedMap<K,V>

Parameters:

fromKey - low endpoint (inclusive) of the keys in the returned map

Returns:

a view of the portion of this map whose keys are greater than or equal to fromKey

Throws:

ClassCastException - if fromKey is not compatible with this map's comparator (or, if the map has no comparator, if fromKey does not implement Comparable). Implementations may, but are not required to, throw this exception if fromKey cannot be compared to keys currently in the map.

NullPointerException - if fromKey is null and this map does not permit null keys

IllegalArgumentException - if this map itself has a restricted range, and fromKey lies outside the bounds of the range

descendingMap

ConcurrentNavigableMap<K,V> descendingMap()

Returns a reverse order view of the mappings contained in this map. The descending map is backed by this map, so changes to the map are reflected in the descending map, and vice-versa.

The returned map has an ordering equivalent to Collections.reverseOrder(comparator()). The expression m.descendingMap().descendingMap() returns a view of m essentially equivalent to m.

Specified by:

descendingMap in interface NavigableMap<K,V>

Returns:

a reverse order view of this map

navigableKeySet

NavigableSet<K> navigableKeySet()

Returns a NavigableSet view of the keys contained in this map. The set's iterator returns the keys in ascending order. The set is backed by the map, so changes to the map are reflected in the set, and vice-versa. The set supports element removal, which removes the corresponding mapping from the map, via the Iterator.remove, Set.remove, removeAll, retainAll, and clear operations. It does not support the add or addAll operations.

The view's iterators and spliterators are *weakly consistent*.

Specified by:

navigableKeySet in interface NavigableMap<K,V>

Returns:

a navigable set view of the keys in this map

keySet

NavigableSet<K> keySet()

Returns a `NavigableSet` view of the keys contained in this map. The set's iterator returns the keys in ascending order. The set is backed by the map, so changes to the map are reflected in the set, and vice-versa. The set supports element removal, which removes the corresponding mapping from the map, via the `Iterator.remove`, `Set.remove`, `removeAll`, `retainAll`, and `clear` operations. It does not support the `add` or `addAll` operations.

The view's iterators and spliterators are *weakly consistent*.

This method is equivalent to method `navigableKeySet`.

Specified by:

`keySet` in interface `Map<K,V>`

Specified by:

`keySet` in interface `SortedMap<K,V>`

Returns:

a navigable set view of the keys in this map

descendingKeySet

`NavigableSet<K> descendingKeySet()`

Returns a reverse order `NavigableSet` view of the keys contained in this map. The set's iterator returns the keys in descending order. The set is backed by the map, so changes to the map are reflected in the set, and vice-versa. The set supports element removal, which removes the corresponding mapping from the map, via the `Iterator.remove`, `Set.remove`, `removeAll`, `retainAll`, and `clear` operations. It does not support the `add` or `addAll` operations.

The view's iterators and spliterators are *weakly consistent*.

Specified by:

`descendingKeySet` in interface `NavigableMap<K,V>`

Returns:

```
a reverse order navigable set view of the keys in this map
```


Submit a bug or feature

For further API reference and developer documentation, see Java SE Documentation. That documentation contains more detailed, developer-targeted descriptions, with conceptual overviews, definitions of terms, workarounds, and working code examples.

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